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(54) **SEAL FOR AN ELECTRONIC DEVICE**

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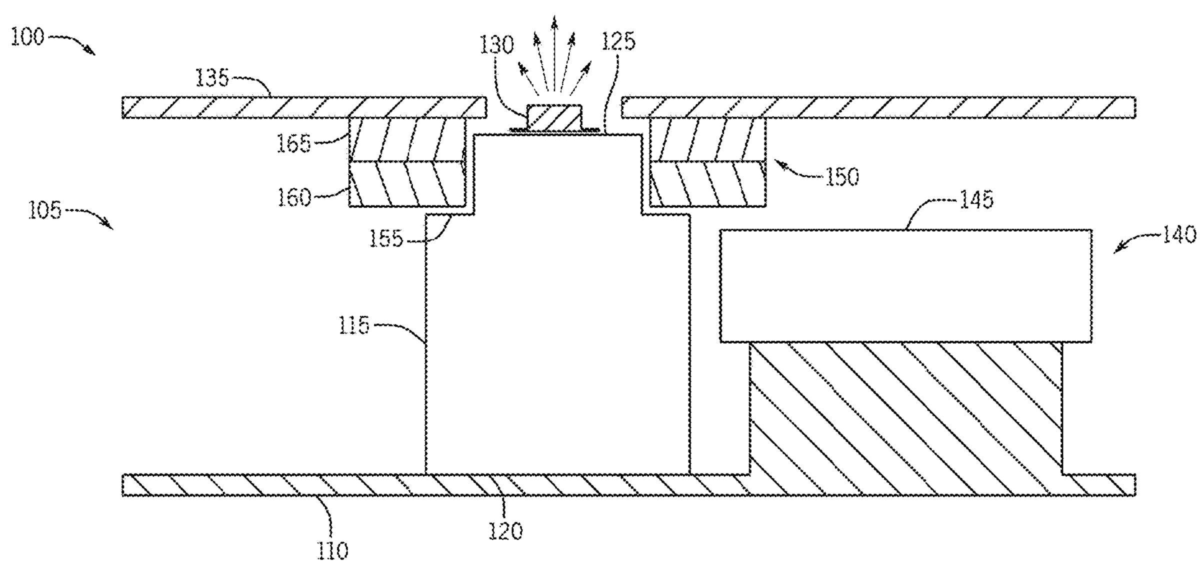
(57) **ABSTRACT**

An electronic device includes housing having a first wall and a second wall, the first and second walls defining an enclosure. A base support is disposed in the enclosure and defining a first surface and a second surface opposite the first surface. The first surface is coupled to the first wall. A light emitting component is coupled to a second surface of the base support. A seal surrounding the light emitting component is disposed between the base support and the second wall. The second wall is disposed over the base support.



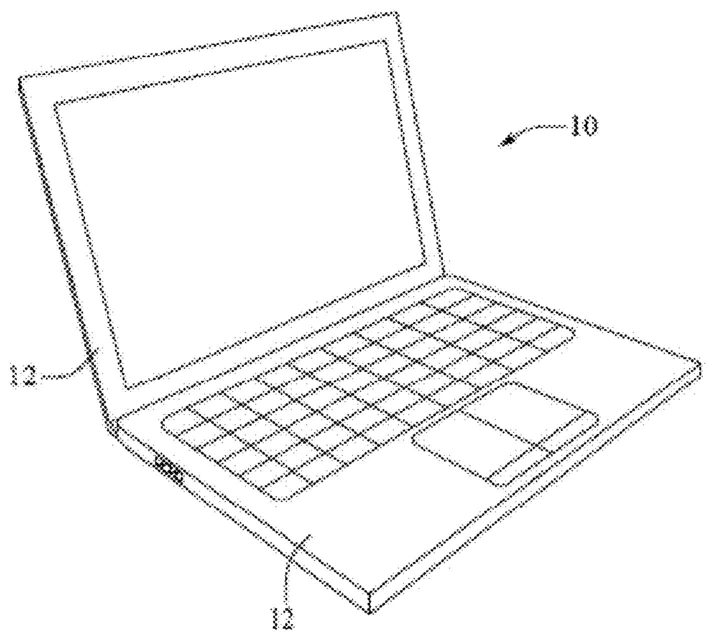


FIG. 1

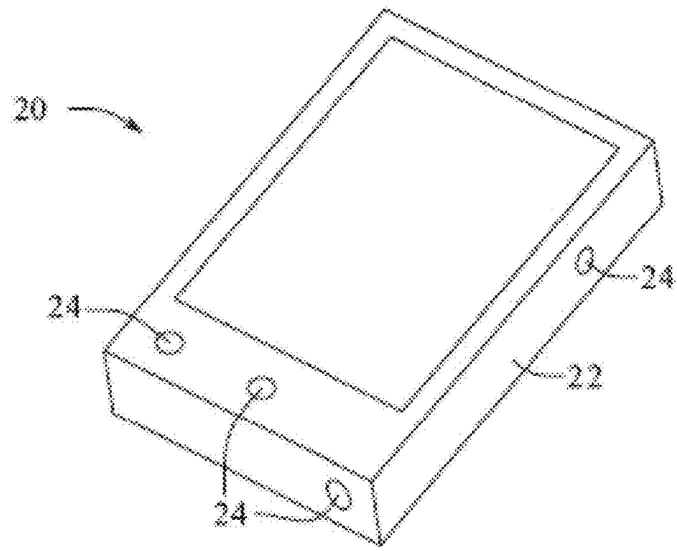


FIG. 2

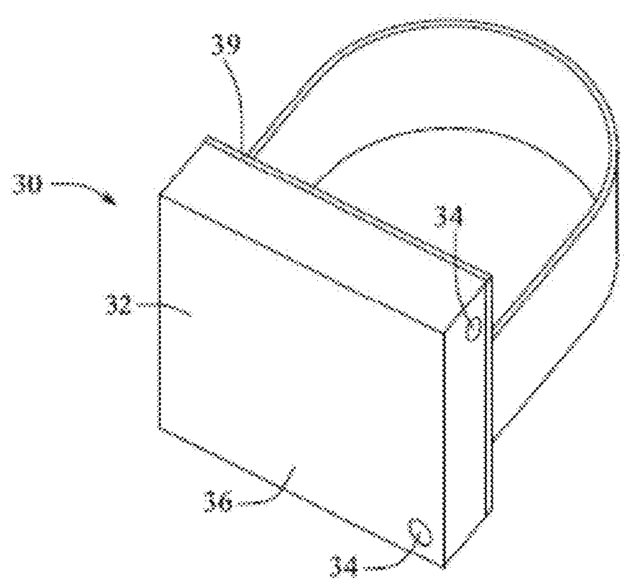


FIG. 3

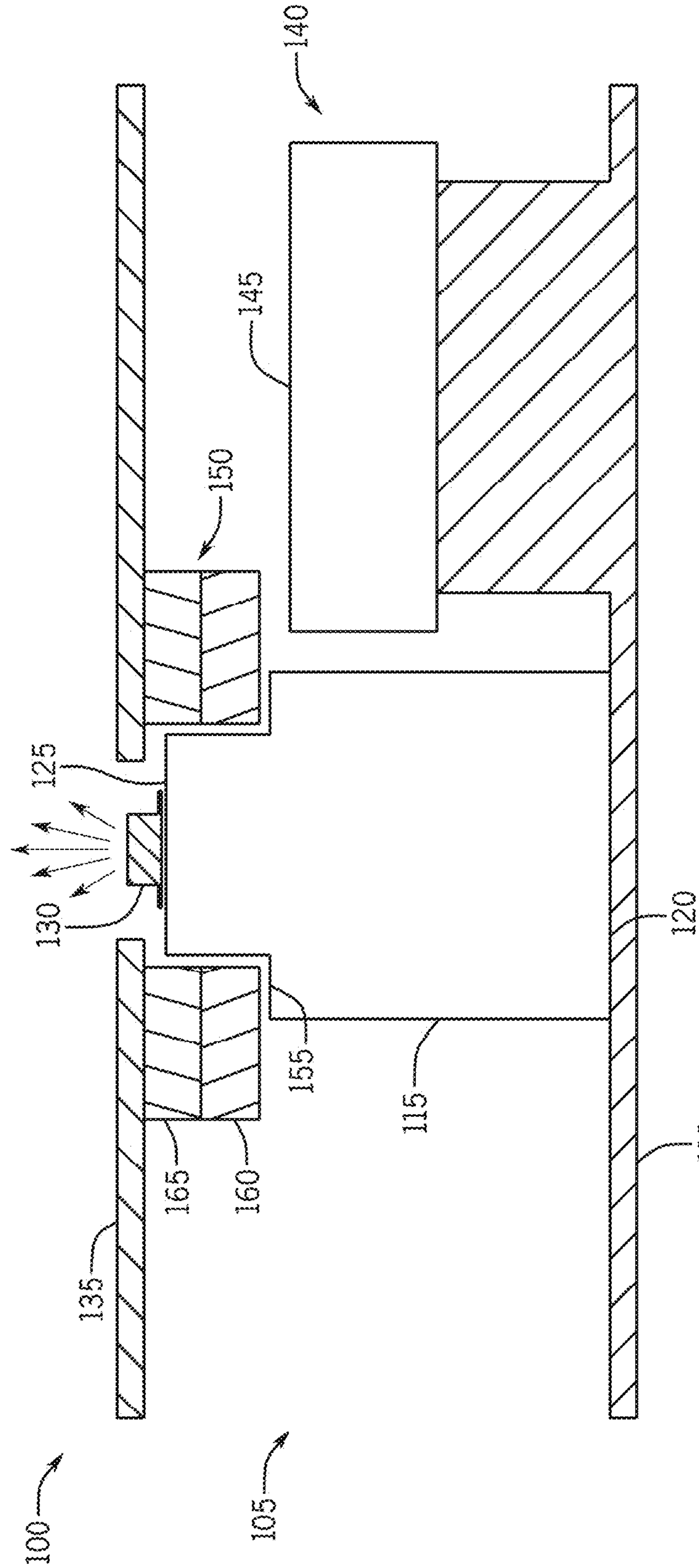


FIG.4

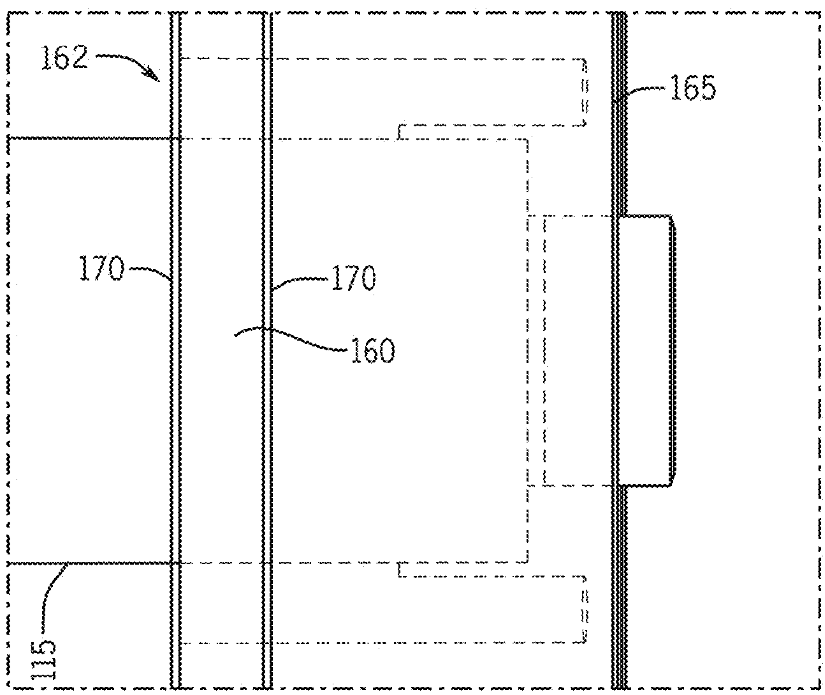
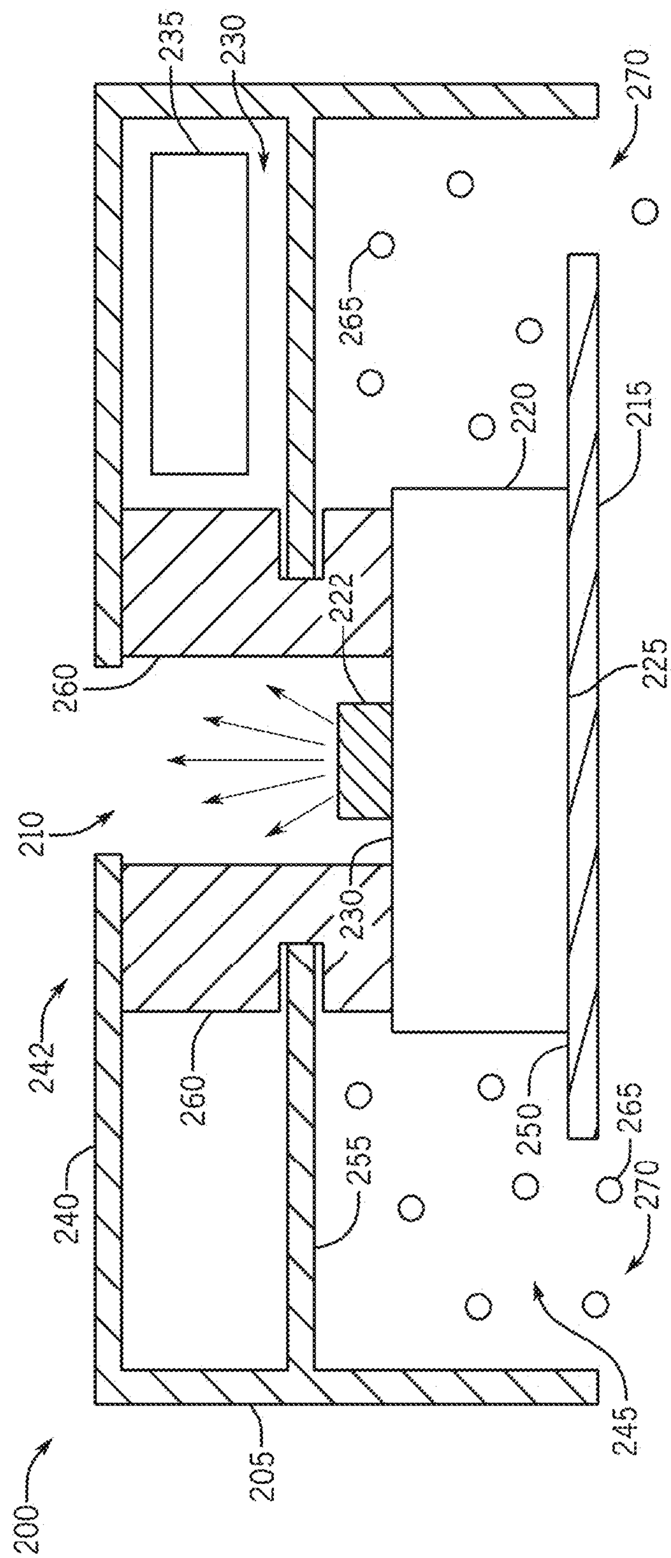


FIG. 5


235
230
270
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222
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FIG.6

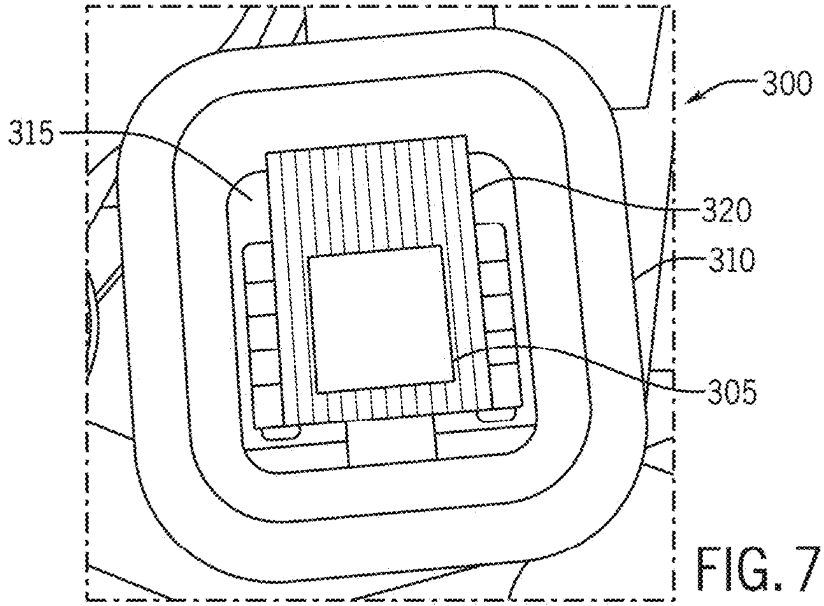

FIG. 7A

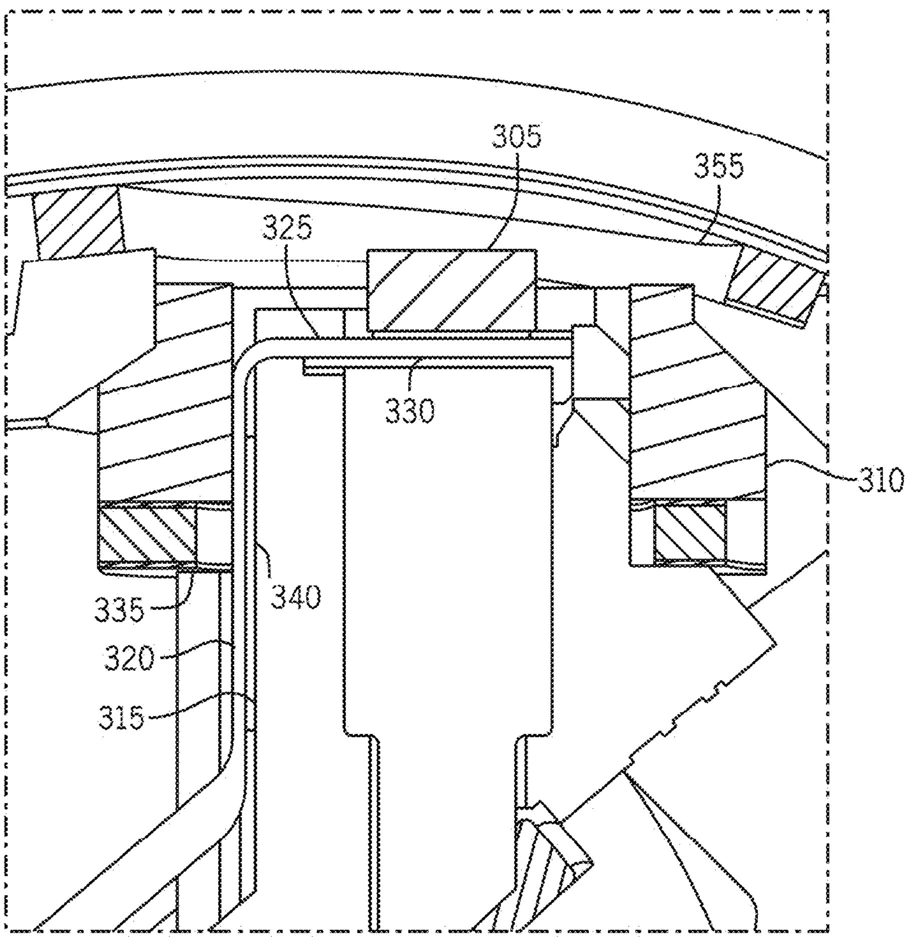

FIG. 7B

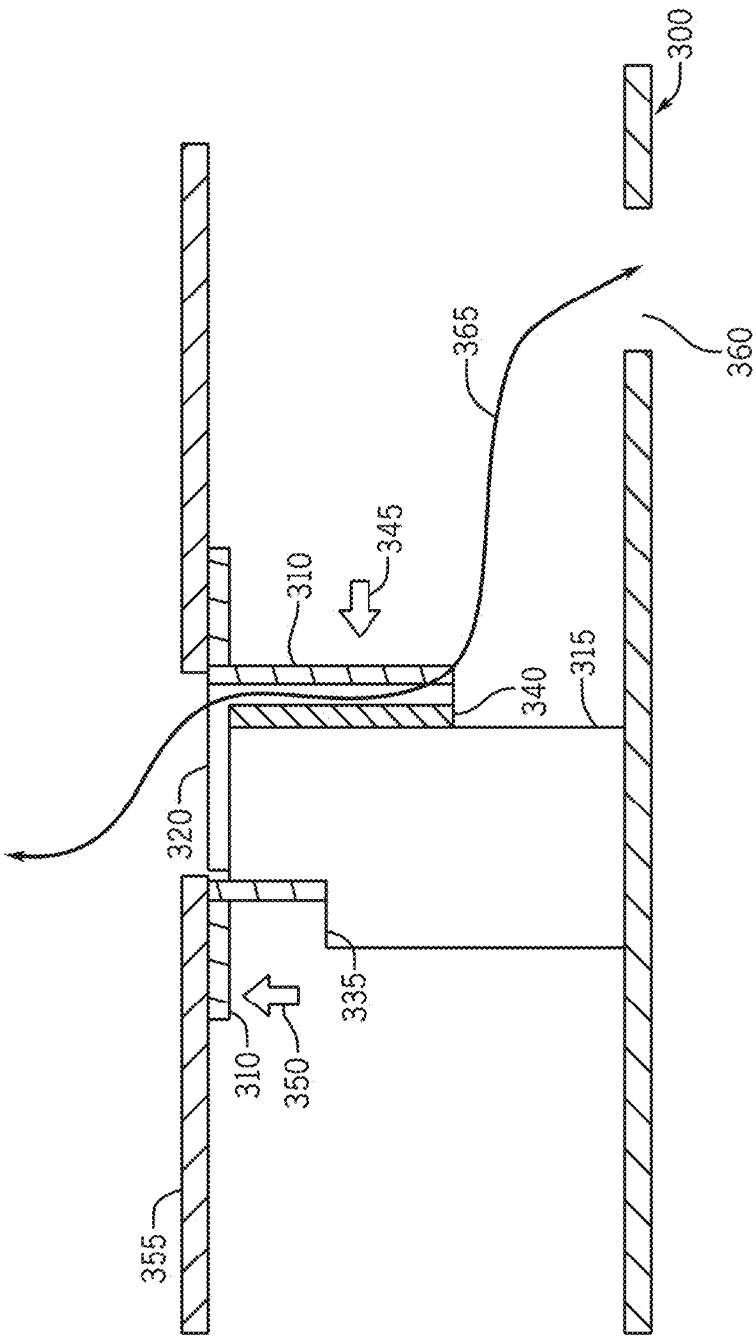

FIG. 7C

SEAL FOR AN ELECTRONIC DEVICE

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a National Stage filing based off of PCT Application No. PCT/US2023/02420, filed 1 Jun. 2023, and entitled "SEAL FOR AN ELECTRONIC DEVICE" which claims priority to U.S. Provisional Patent Application No. 63/365,759, filed 2 Jun. 2022, and entitled "SEAL FOR AN ELECTRONIC DEVICE," the entire disclosure of which is hereby incorporated by reference.

FIELD

[0002] The described embodiments relate generally to the protection of electronic devices and components therein. More particularly, the present embodiments relate to dust seals that separate electrical components and are configured to aid assembly.

BACKGROUND

[0003] Electronic devices can include internal components that communicate, interface, or interact with the environment external to the device in some way. Often, these internal components can be negatively affected by the external environment, including interference from external light and dust. Such components can also interfere with each other and negatively affect their intended functions. Further, the components are susceptible to degradation due to dust and debris or light interference.

[0004] However, traditional electronic devices and components are not designed adequately to withstand external forces while protecting the internal components from both dust damage and light interference. In addition, traditional designs attempting to mitigate such interferences make it difficult or expensive to assemble the electronic device. Furthermore, traditional designs tend to be heavy and can be bulky or lack aesthetically and tactilely pleasing design features.

SUMMARY

[0005] An electronic device can include a housing including a first wall and a second wall, the first and second walls defining an enclosure. The device can also include a base support disposed in the enclosure and defining a first surface and a second surface opposite the first surface. The first surface can be coupled to the first wall. A light emitting component can be coupled to a second surface of the base support. The electronic device can further include a seal surrounding the light emitting component disposed between the base support and the second wall. In some examples, the second wall can be disposed over the base support. In some examples, an electronic component can be disposed between the first wall and the second wall, and the seal can separate the electronic component from the light emitting component. The base support can include a ledge. In some examples, the seal can surround an upper portion of the base support and can contact the ledge. The seal and the base support can be engaged in an interference fit.

[0006] In some examples, the seal can include a plastic shim that contacts the base support and a foam material coupled to the plastic shim with a pressure sensitive adhesive. The foam material can be compressed between the plastic shim and the second wall. In other examples, the seal can include a silicone material. The seal can obstruct light having wavelengths between about 10^{-8} m to about 10^{-3} m and inhibit dust and/or debris from traversing between the base support and the second wall. In some examples, the seal can be die cut.

[0007] In some examples, a head mountable electronic device can include a housing having an internal wall defining an interior surface and an external wall defining an external surface. The head mountable electronic device can further include a first enclosure defined by the internal wall and the external wall, a second enclosure defined by the internal wall and the external wall, and a third enclosure defined by the internal wall. A light emitting component can be disposed in the first enclosure and a light sensitive component can be disposed in the second enclosure. A metal or a plastic base support can be disposed in the third enclosure, the base support including a top surface. The electronic device can further include a seal coupled to the top portion of the base support. The seal can separate the first enclosure, the second enclosure, and the third enclosure. In some examples, the seal can be impenetrable to dust, debris, and light. In some examples, the light emitting component can include a light emitting diode. In some examples, the light sensitive component can include at least one of a camera, an ambient light sensor, or a flicker sensor.

[0008] In some examples, the seal can include a compressible foam. In some examples, the seal can extend from the base support and through the internal wall to the exterior surface of the housing. In some examples, the electronic device can further include a flexible printed circuit board disposed between the seal and the base support, the seal retaining the flexible printed circuit board adjacent the base support. In some examples, at least one of the interior surface and external surface of the housing can include an air vent. In some examples, the light emitting component can be coupled to a top surface of the base support. The base support can include a ledge. In some examples, the seal can surround an upper portion of the base support, the seal having a plastic shim contacting the ledge.

[0009] In some examples, an electronic device can include a dust seal coupled to a base support. The dust seal can couple to the support in an interference fit. The electronic device can also include a flexible printed circuit board disposed between the dust seal and the base support. In some examples, the dust seal can retain the flexible printed circuit board adjacent the base support. In some examples, the flexible printed circuit board can be coupled to a light emitting device mounted to the base support. In some examples, the dust seal can encircle the light emitting device and the dust seal can be impenetrable to light and dust. In some examples, the dust seal can include a plastic shim and a foam portion, the plastic shim having a bottom surface coupled to the base support with a pressure sensitive adhesive and a top surface coupled to the foam portion with the pressure sensitive adhesive. The foam portion can include a compressible open or closed cell foam. In some examples, the base support can include a ledge and the dust seal can surround an upper portion of the base support and the plastic shim can contact the ledge.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements, and in which:

[0011] FIG. **1** shows an example of an electronic device;

[0012] FIG. **2** shows another example of an electronic device;

[0013] FIG. **3** shows yet another example of an electronic device;

[0014] FIG. **4** shows a cross-sectional view of a portion of an electronic device, according to an example;

[0015] FIG. **5** shows a cross-sectional view of a seal, according to an example;

[0016] FIG. **6** shows a cross-sectional view of an electronic device, according to an example;

[0017] FIG. **7**A shows a top perspective view of an electronic component and a seal, according to an example;

[0018] FIG. **7**B shows a cross-sectional view of a portion of an electronic device, according to an example; and

[0019] FIG. **7**C shows a cross-sectional view of a portion of an electronic device, according to an example.

DETAILED DESCRIPTION

[0020] Reference will now be made in detail to representative examples illustrated in the accompanying drawings. It should be understood that the following descriptions are not intended to limit the examples to one preferred example. To the contrary, it is intended to cover alternatives, modifications, and equivalents as can be included within the spirit and scope of the described examples as defined by the appended claims.

[0021] The following disclosure relates to seals, electronic components protected by the seals, and arrangements for electronic components within electronic devices. Examples of electronic devices described herein include portable electronic devices and personal computing devices designed to be used in a variety of environments. For example, many people carry mobile phones on their person throughout the day and night, including during outdoor activities or inside an office or home. Electronic watches are also worn and used in many environments including outdoors while recreating, submerged in water, in the rain, under the sun, and so forth. A more recent technological development is the use of virtual reality and/or alternate reality (VR/AR) headsets that can be worn and used inside or outside and during the day or night.

[0022] Any of these examples of electronic devices can include a number of components that interact with the external environment. These components can include sensors, antennas, cameras, or other similar electronic components. In the example of the AR/VR headset given above, one or more sensors can include tracking sensors, ambient light sensors, and/or flicker sensors. Also, one or more outward facing lights, such as light-emitting diodes (LEDs) can be placed and tracked with an external camera to develop accurate positioning of the VR/AR headset and/or other components.

[0023] The electronic devices described in the present disclosure include components and arrangements configured to protect the LEDs, antennas, sensors, and/or other components communicating with the external environment noted above, from light and debris that may otherwise harm or negatively affect the components. More specifically, the present disclosure describes seals and electronic components that ensure internal components can sufficiently function without interference from an environment external to the electronic device, as needed, without the risk of damage to the internal component. Such interferences can include light, debris, or moisture. Such internal components can include, but are not limited to, cameras, LEDs, ambient light sensors, flicker sensors, and other sensors, and the like. These and other internal components can be disposed adjacent to, or aligned below, one or more apertures/ports of the electronic device, with a seal disposed between the aperture/vent and the internal electronic component. In this way, the seals described herein can act as a barrier between the internal component and the external environment or the internal component and other internal components while also enabling the proper functioning of the internal component.

[0024] In a particular example, an electronic device includes a housing that defines an internal volume and an aperture defined by the housing. An electronic component can be disposed within the internal volume and a seal can be disposed between the aperture and the electronic component, the seal being configured to protect the electronic component within the housing.

[0025] In a particular example, an electronic component can include a light emitting component coupled to a base support. The electronic device can include other electronic components that are light sensitive. The seal can be disposed and configured between the light emitting component and light sensitive component to prevent and/or minimize light interference.

[0026] The seals described herein obstruct light having wavelengths in the visible range, as well as those having wavelengths extending into to ultraviolet and infrared ranges. The seals also inhibit dust and/or debris from traversing into the housing from apertures and/or any vents disposed on the housing of the electronic device. The seals described herein are also compressible to aid in assembly of the electronic device. In addition, the seals and assemblies described herein are durable and resistant to damage. The seal can be die cut and the components of the electronic device can be made of materials intended to minimize the weight of the components and/or the electronic device.

[0027] For example, a seal described herein can include an open or closed cell foam. The seal can engage a bracket or base support of the electronic device in an interference fit. In some examples, the seal can include a seal configured to surround the light emitting component. Because the light emitting component is mounted to an end of the base support in some examples, the seal can be die cut to have a precise interference fit with the base support to separate the light emitting component from other electronic components disposed in the electronic device and to facilitate a top-down assembly orientation.

[0028] These and other examples are discussed below with reference to FIGS. **1**-**7**C. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these Figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[0029] In the present disclosure, any example of an electronic device that incorporates the housing, enclosures, electronic components, seals, and configurations described herein can incorporate the present teachings and methods. The figures herein are for illustrative purposes only to show an example implementation of the device and the electronic components and configurations of the present disclosure. The seals, base supports, housing, and seal/enclosure configurations can be implemented in any number of electronic devices. Examples of electronic components, seal components, and device configurations disclosed herein can be included in any number of electronic devices, including but not limited to desktop computers, laptop computers, tablets, smartphones, smart speakers, wearable electronic devices such as fitness trackers, smart watches, head-mountable-display devices or other alternate/virtual reality devices, and so forth.

[0030] FIG. **1** illustrates an example of an electronic computing device **10** having a compact design. The device **10** is a laptop computer having multiple functions, including visual and audio outputs, internet connectivity, electronic communication capabilities, and computing functionalities. These functions can be enabled by a multitude of components, including processors, antennas, display components, screens, and lights, input components such as keyboards and touchpads, and so forth. Many of these electronic components can include sensors and/or light-emitting components that can interfere with nearby components of the device **10**. These components can be isolated by the present exemplary systems and methods for proper operation.

[0031] In one example, the device **10** can include a housing **12** or multiple housings defining an internal volume. The sensors and/or light-emitting components can be disposed within the internal volume or outside the internal volume. In on example, the components can communicate with the ambient environment through the housing. For example, the light-emitting component can be an LED disposed at or on the housing **12**, or within the housing **12** and directing light through an aperture defined by the housing **12**. The sensor components and systems described herein can be disposed within the internal volume defined by the housing **12**. Another exemplary electronic device is detailed below with reference to FIG. **2**.

[0032] FIG. **2** shows another example of an electronic device **20**, including a smartphone having a housing **22** defining an internal volume and an external surface of the device **20**. The smartphone **20** can also include a number of electronic components, including display/touch screens, and/or sensors and one or more lights **24** can be disposed at various locations on, above, or within the housing **22** as shown. In one example, the lights **24** can be notification lights or visual icons disposed within the internal volume or outside the internal volume defined by the housing **22** of the device **20**. In on example, the lights **24** can communicate with the ambient environment through the housing **22**. For example, one or more of the lights **24** can be an LED disposed at or on the housing **22**, or within the housing **22**, and directing light through an aperture defined by the housing **22**.

[0033] In the illustrated example of FIG. **2**, the lights **24** can include LEDs that can interfere with light-sensitive components. The light-emitting and light sensitive components and systems described herein can be included in the smartphone **20** while being isolated from each other to prevent degradation and interference of the light sensitive components by the light-emitting components. Another exemplary electronic device is detailed below with reference to FIG. **3**.

[0034] FIG. **3** shows another example of an electronic device, including a head-mountable display **30** having a display portion **36**. The display portion **36** can include a housing **32** defining an internal volume and one or more light emitting and/or light sensitive electronic components, including outward facing lights **34**. The display portion **36** can also include a display screen **39** or other display component projecting light toward the user's eyes when the user dons the head-mountable display **30**. The lights **34** can include one or more LEDs. The lights **34** and/or light sensitive components can be disposed at various locations on, above, or within the housing **32** as shown. In one example, the lights **34** can be notification lights or visual icons disposed within the internal volume or outside the internal volume defined by the housing **32** of the head-mountable display **30**. In one example, the lights **34** can communicate with the ambient environment through the housing **32**. Light sensitive components can include ambient light sensors, cameras, and/or flicker sensors. One or more of the lights **34** can be an LED disposed at or on the housing **32**, or within the housing **32** and directing light through an aperture defined by the housing **32**. In other embodiments, the lights **34** can be infrared or other non-visible lights used to aid sensors or other components of the head-mountable display **30** by illuminating objects with light of a known frequency.

[0035] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **1-3** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **1-3**. Further details of the present sealing systems and methods are provided below with reference to FIG. **4**.

[0036] FIG. **4** illustrates a cross-section of an electronic device **100**. The electronic device can include any of the electronic devices described above. The electronic device can include a housing **105**. The housing can include a first wall **110**. In some examples, the first wall **110** can include an exterior wall of the electronic device **100** or can include a wall that separate chambers or compartments within electronic device **100**. The housing **105** can include a base support **115** coupled to the first wall **110**. The base support **115** can include a base bracket or support piece for the electronic device **100** and/or electronic components disposed within the housing **105**. In some examples, the base support **115** can include one or more stiff metal materials, including steel, stainless steel, magnesium, and/or titanium. In some examples, the base support **115** can include plastic and/or plastic components to reduce weight of the electronic device **100**. The base support **115** can be cylindrical shaped or can include a cubic shape and may be a solid material or hollow to reduce weight of the electronic device **100**. The base support **115** can include a first surface **120** coupled to the first wall **110** by adhesive or any non-limiting fastening method and/or device. An electronic component can be coupled to a second surface **125** of the base support **115**. The second surface **125** of the base support **115** can be configured opposite the first surface **120**. The electronic component can include a light emitting component **130** (e.g., a light emitting diode or LED).

[0037] The electronic device **100** can further include a second wall **135** disposed over the base support **115**. The first wall **110** and the second wall **135** can form an enclosure **140** within the housing **105** and the base support **115** can be disposed within the enclosure **140**. In some examples, an electronic component **145** can also be disposed within the enclosure **140**. The electronic component **145** can include a light sensitive component, in some examples, subject to light interference from the light emitting component **130**. The electronic component **145** can be mounted to a second base support or can be coupled to either the first wall **110** and/or the second wall **135** directly.

[0038] The electronic device **100** can further include a seal **150** surrounding the light emitting component **130**. The seal **150** can be disposed between the base support **115** and the second wall **135**. In some examples, the seal **150** separates the electronic component **145** from the light emitting component **130**. The seal **150** can be configured to obstruct light having wavelengths between about 10^{-8} m and about 10^{-3} m. In other words, the seal **150** can obstruct light in the visible region, the ultraviolet region, and the infrared region. The seal **150** can also inhibit dust and/or debris from traversing between the base support **115** and the second wall **135**. In some examples, the seal **150** surrounds the light emitting component **130** on all sides. As such, the seal **150** engages the base support **115** either around a perimeter of an upper portion of the base support **115** or the second surface **125** of the base support **115**. In some examples, the second surface **125** includes the same surface having the light emitting component **130** disposed thereon.

[0039] In some examples, the base support **115** defines a ledge **155**. The seal **150** can surround an upper portion of the base support **115** and contact or engage an upper surface of the ledge **155**. As such, the seal **150** and the base support **115** can be engaged in an interference fit. For example, the radial seal **150** can be disposed in a press fit or friction fit between the base support **115** and the second wall **135**. The seal **150** and the base support **115** can be held together by friction after the housing **105** and other components are assembled. For example, the seal **150** can be compressed between the second wall **135** and the second surface **125** or the ledge **155** of the base support **115** to separate or seal the light emitting component **130** from the electronic component **145**. In some examples, the seal **150** can separate and/or seal off the enclosure **140** from an area or environment outside the housing **105** or external to the electronic device **100**.

[0040] In some examples, the seal **150** can be die cut. The seal **150** can be cut into a precise shape by a metal "die" during production. Because of the precise shape (e.g., radial) and dimensions of the seal **150** to properly seal against light, dust, and other debris and to fit properly against the second wall **135** and the base support **115**, including the ledge **155** defined thereby, the die cutting process can provide the exact shape and properties of the seal **150** for proper assembly of the electronic device **100**.

[0041] In some examples, the seal **150** can include a silicone material. The silicone seal **150** can be manufactured inexpensively and when shaped appropriately, provide the required compressibility and interference fit to the base support **115**. In some examples, the silicone seal **150** can include a foaming agent and/or be dyed to obstruct light from passing through from either the external environment and/or the light emitting component **130** to the electronic component **145** within the enclosure **140**. The seal **150** can include portions formed of different materials. In some examples, the seal **150** can include a plastic shim **160**. The plastic shim **160** can be included as a component of the seal **150** to reduce weight of the seal **150**, and thus, can reduce the weight of the electronic device **100**. In some examples, the plastic shim **160** can contact the base support **115**. In some examples, the plastic shim **160** can contact the ledge **155** and/or the upper portion of the base support **115**. The plastic shim **160** can also be impenetrable or resistant to light, dust, and/or debris. The seal can further include a foam material **165** coupled to the plastic shim **160**.

[0042] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **4** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **4**. Further details of the seal are provided below with reference to FIG. **5**.

[0043] Referring now to FIG. **5**, a cross-section of the seal **150** illustrated in FIG. **4** is shown. In some examples, the plastic shim **160** can be coupled to the foam material **165** with a pressure sensitive adhesive **170**, or any suitable adhesive. The plastic shim **160** can form a precision seal with the base support **115** and can be less expensive to manufacture than the silicone and/or a seal including only foam material. The foam material **165** can be compressed between the plastic shim **160** and the second wall **135**, as shown in FIG. **4**. The plastic shim **160** can include a bottom surface **162** coupled to the base support **115**. The plastic shim **160** can be joined to the base support **115** with the pressure sensitive adhesive **170**. A top surface of the plastic shim **160** can also include a pressure sensitive adhesive **170** to couple the plastic shim **160** to the foam material **165**. In some examples, the foam material **165** can include a compressible open or closed cell foam to inhibit light, dust, and/or debris.

[0044] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **5** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **5**. Additional configurations of the seals described herein are provided below with reference to FIGS. **6**-**7**C.

[0045] FIG. **6** illustrates an electronic device **200** having a housing **205** and electronic components therein. The electronic device **200** can include a head mountable device (e.g., device **30**). In some examples, the electronic device **200** can include a first enclosure **210** having a light emitting component **222** disposed therein. The first enclosure can be defined, at least in part, by the housing **205**, an internal wall **255**, and an external wall **240** defining an external surface **242**. The housing **205** can include a first wall **215**. In some examples, the first wall **215** can include an exterior wall of the electronic device **200** or can include a wall that separate chambers or compartments within the electronic device **200**. The housing **205** can include a base support **220** coupled to the first wall **215**. The base support **220** can include a first surface **225** coupled to the first wall **215** by an adhesive or any non-limiting fastening method and/or device. An electronic component can be coupled to a second surface **230** of the base support **220**. The second surface **230** of the base support **220** can be opposite the first surface **225**. The light emitting component **222** can include a light emitting diode (LED).

[0046] The housing **205** can further include a second enclosure **230**. The second enclosure **230** can include at least one light sensitive component **235** disposed therein. The second enclosure **230** can be defined by the housing **205**, including various internal and external walls. In some examples, the light sensitive component **235** can include an optically sensitive device. In some examples, the light sensitive component **235** can include at least one of a camera, an ambient light sensor, or a flicker sensor. In some examples, the second enclosure **230** can include an exterior surface **240** of the housing **205**. In other words, one of the walls of the second enclosure **230** includes the exterior surface **240** of the housing **205**.

[0047] In some examples, the housing **205** can further define a third enclosure **245**. The base support **220** can be disposed within the third enclosure **245**. In some examples, the base support **220** can include a metal or plastic material. In some examples, the third enclosure **245** can include an interior surface **250** of the housing **205**. In other words, one of the walls of the third enclosure **245** includes the interior surface **250** of the housing **205**. In some examples, the housing **205** of the electronic device **200** includes an internal wall **255** separating at least a portion of the second enclosure **230** and the third enclosure **245**. In some examples, a seal **260** is included. The seal **260** can be coupled to a top portion or the second surface **230** of the base support **220**. The seal **260** can separate the first enclosure **210**, the second enclosure **230**, and the third enclosure **245**. In other words, the seal **260** includes at least a portion of a wall or divider that separates the first enclosure **210**, the second enclosure **230**, and the third enclosure **245** from each other. In some examples, the seal **260** is impenetrable to dust, debris, and/or light. The seal can extend from the second surface **230** of the base support **220** past the internal wall **255** and press fit to the exterior surface **240** of the housing **205**. In such a configuration as shown in FIG. **6**, the seal **260** functions to both protect the light sensitive component **235** from the light emitting component **222** and protects both the light sensitive component **235** and the light emitting component **222** from debris and/or dust **265** present either outside the housing **205** or within the third enclosure **245**.

[0048] In some examples, the dust **265** can ingress into the third enclosure **245** through an air vent **270** or other aperture. The seal **260** prevents or substantially prevents the dust **265** or other debris, such as, but in no way limited to, water, dirt, and smoke, from passing through to the first enclosure **210** or second enclosure **230**, which can include sensitive electronic components. The air vent **270** can be included on the electronic device **200** for temperature control and/or heat dissipation. In some examples, the air vent **270** can be included for comfort and/or safety for the user of the electronic device **200**. According to one example, the air vent **270** can include a perforated material to cover the air vent **270** or other apertures in the electronic device. The size, locations, and number of perforations extending through such a material can vary from one example to another. Such perforations can include machined, laser cut, or otherwise manufactured openings defined by and extending through the material. Such openings can be sized and arranged to prevent a certain size particle from the external environment from passing through the air vent **270**. Such openings can also be sized and include other types of filters to prevent moisture or other harmful debris from passing through the air vent **270**.

[0049] In some examples, the seal **260** can be cut and/or shaped to seal around the internal wall **255** and between the second surface **230** of the base support **220** and the exterior surface **240** of the housing **205**. The seal **260** can include a compressible foam and can compress in an interference or press fit between the exterior surface **240** of the housing **205** and a top portion or the second surface **230** of the base support **220**. The seal **260** can include an open cell polyurethane foam, which is lightweight and more compressible compared to a closed cell foam. In other examples, the seal **260** can include a silicone material. The seal **260** can include a reticulated Polyurethane, PVC/Nitrile, Ethylene Propylene Diene Monomer (EPDM) rubber, or other suitable material. The materials which can be used to produce the closed cell foam for the seal **260** can vary greatly from ethylene-vinyl acetate (EVA), polyethylene, polystyrene, rubber to polypropylene etc. The closed cell foam can include trapped gas bubbles which are formed during the expansion and cure of the foam. The bubbles are permanently locked to a place, as the trapped gas is very efficient in increasing the insulation capability of the foam. The foam which is formed is strong and is usually of a greater density than open cell foam, which enables the gas bubbles to lock into place. The nature of the foam enables it to be vapor retardant and resist liquid water. In some examples, the materials for the seal can be open celled, but can include a sufficiently tortuous path so as to restrict or eliminate the passage of contaminants between enclosures.

[0050] In some examples, the top portion of the base support **220** can include a ledge (e.g., ledge **155** shown in FIG. **4**). The seal **260** can surround an upper portion of the base support **220** and contact the ledge. In some examples, the seal **260** can further include a plastic shim (e.g., plastic shim **160**) that contacts the ledge. The seal **260** can fit tightly to the base support **220**. The tight fit can function not only to seal the enclosures within the housing, but also to control and/or retain electronic components in place, which can improve the design of the electronic device **200**, minimize noise and/or vibrations, and reduce manufacturing costs.

[0051] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **6**. Further details of an electronic device incorporating the present arrangements is provided below with reference to FIGS. **7**A-**7**C.

[0052] FIGS. **7**A-**7**C illustrate an electronic device **300**. FIG. **7**A shows a top perspective view of an electronic component **305** and a dust seal **310** surrounding the electronic component **305** within the electronic device **300**, according to one example. In some examples, the dust seal **310** can be coupled to a base support **315**. The dust seal **310** can couple to the base support **315** via an interference fit. In some examples, the dust seal **310** can be adhered to the base support **315** with an adhesive (e.g., pressure sensitive adhesive **170** shown in FIG. **5**). In some examples, the electronic component **305** can include a light emitting component (e.g., light emitting component **222** shown in FIG. **6**). The electronic device **300** can further include a flexible printed circuit board or flexible printed circuit (flex PCB) **320**. At least a portion of the flex PCB **320** can be disposed between the dust seal **310** and the base support **315**. In some examples, a flex PCB can include an arrangement of printed circuitry and/or components that utilize flexible materials with a flexible overlay. A flexible printed circuit board can include a metallic layer of traces, usually copper, bonded to a dielectric layer. Thickness of the metallic layer can be very thin (<0.0001") to very thick (>0.010") and the dielectric thickness can also vary. The flex PCB can be included in the electronic device **300** to reduce the assembly process and improve reliability. The flex PCB can be used as a connector, power supply, and also as full circuits assembled with components such as electronic component **305**. A benefit of including the flex PCB **320** in the electronic device **300** is connecting the electronic component **305** to other components and/or a power source (not shown) and having the flex PCB **320** disposed between the dust seal **310** and the base support **315**. In some examples, the dust seal **310** retains the flex PCB adjacent to the base support **315**.

[0053] FIG. **7**B shows a cross-sectional view of a portion of electronic device **300**. The flex PCB **320** can be coupled to the electronic component **305**. In some examples, electronic component **305** can include a light emitting device mounted to the base support **315**. In some examples, the light emitting device can be coupled to a top surface **325** of the flex PCB **320** and a bottom surface **330** of the flex PCB **320** can be coupled to the base support **315**. The dust seal **310** can encircle the light emitting device. In some examples, the dust seal **310** is impenetrable or resistant to light, dust, and/or debris. FIG. **7**B shows the base support **315** having a ledge **335**. The dust seal **310** can extend from the upper portion of the base support **315** adjacent to the base support **315** and past the dust seal **310**. In some examples, the ledge **335** can include a cut-out portion. The cut-out portion can have the width of the flex PCB to minimize folding of the PCB and retain the flex PCB **320** adjacent to the base support **315**. Referring to FIG. **7**C, the flex PCB **320** can further include an adhesive **340** on the bottom surface **330** to retain the flex PCB **320** adjacent to the base support **315**. FIG. **7**C shows components of the dust seal, each component configured to retain the flex PCB **320** and/or prevent dust and/or debris ingress into the electronic device.

[0054] In some examples, the dust seal **310** can act as a radial seal to apply a radial force **345** to the flex PCB **320** to retain the flex PCB **320** adjacent the base support **315**. In other words, the dust seal **310** prevents ingress path and aids adhesive **340** in flex management by applying compressive normal force that keeps the flex PCB **320** from delaminating from the base support **315**. The dust seal **310** can also act as a compressive seal to apply a compressive force **350** to compress the dust seal **310** between a housing wall **355** and the base support **315**. As such, the dust seal **310** can include a compressible foam. The dust seal **310** can be coupled to the base support **315** and engaged to the base support and the housing wall **355** in an interference fit. As described in other examples above, the housing wall **355** of the electronic device **300** can include a vent **360** and/or other apertures. FIG. **7**C includes the dust ingress route **365** that the dust seal **310** is configured to prevent. Because the dust seal **310** includes a compressible foam and is engaged in a compressive seal and a radial seal, the dust ingress route **365** is closed off, the dust seal **310** being impenetrable to dust and/or debris. In some examples, the dust seal **310** is also impenetrable to light.

[0055] In some examples, the components described above for the electronic device examples (e.g., electronic device **100**, **200**, and/or **300**) can aid in the assembly of the electronic device housing (e.g. housing **305**). The assembly process can, in some examples, demand the use of a minimal amount of metal material for the base support **315** to maximize the available distance that can be used between the components. The assembly methods can also demand that the ledge **335** be minimized so that the base support **315** does not interfere with the inclusion/installation of other electronic components (e.g., electronic component **145**) during assembly. In other words, due to the materials used for the base support **315** and/or the shape of the base support **315**, the base support **315** can be installed first, then the other electronic components (e.g., electronic component **145**) can be installed after the base support **315**, followed by the dust seal **310** being placed on or around the base support, and finally the housing wall (e.g., housing wall **355**) can be installed to compress the dust seal **310** in an interference fit between the housing wall **355** and the base support **315**. In other words, the present exemplary sealing systems and methods facilitate a top or front assembly of the device, while ensuring a secure light and/or dust seal between sections.

[0056] Any of the features, components, parts, including the arrangements and configurations thereof shown in FIGS. **7**A-**7**C can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures. Likewise, any of the features, components, parts, including the arrangements and configurations thereof shown in the other figures can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **7**A-**7**C.

[0057] To the extent applicable to the present technology, personal information data may be collected that uniquely identifies or can be used to contact or locate a specific person in order to improve the user experience. Any collection of personal information should be conducted in compliance with well-established privacy policies and practices that meet or exceed industry and governmental requirements for privacy and security. However, the present exemplary systems and methods are fully operable without the collection or use of personal information data.

[0058] Furthermore, the present description incorporates and details a number of examples and alternatives, many of which are not required to perform the present systems and methods. Additionally, elements of the disclosed examples can be combined or varied from the precise forms disclosed herein. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An electronic device, comprising:
- a housing having a first wall and a second wall, the first wall and the second wall defining an enclosure;
- a base support disposed in the enclosure and defining a first surface and a second surface opposite the first surface, the first surface coupled to the first wall;
- a light emitting component coupled to the second surface; and
- a seal surrounding the light emitting component and disposed between the base support and the second wall, the second wall being disposed over the base support.

2. The electronic device of claim **1**, further comprising an electronic component disposed between the first wall and the second wall, the seal separating the electronic component from the light emitting component.

3. The electronic device of claim **1**, wherein:
- the base support comprises a ledge;
- the seal surrounds an upper portion of the base support and contacts the ledge; and
- the seal and the base support are engaged in an interference fit.

4. The electronic device of claim **1**, the seal comprising a plastic shim that contacts the base support and further comprising a foam material connected to the plastic shim by a pressure sensitive adhesive, the foam material compressed between the plastic shim and the second wall.

5. The electronic device of claim **1**, wherein the seal comprises a silicone material.

6. The electronic device of claim **1**, wherein the seal obstructs light having wavelengths between about 10^{-8} m to about 10^{-3} m and inhibits debris from traversing between the base support and the second wall.

7. The electronic device of claim **1**, wherein the seal is die cut.

8. A head mountable electronic device, comprising:
- a housing comprising an internal wall defining an interior surface and an external wall defining an external surface;
- a first enclosure defined by the internal wall and the external wall;
- a second enclosure defined by the internal wall and the external wall;
- a third enclosure defined by the internal wall;
- a light emitting component disposed in first the enclosure;
- a light sensitive component disposed in the second enclosure;
- a base support disposed in the third enclosure, the base support including a top surface; and
- a dust and light seal coupled to the top surface, the seal separating the first enclosure, the second enclosure, and the third enclosure.

9. The electronic device of claim **8**, wherein the light emitting component comprises a light emitting diode.

10. The electronic device of claim **8**, wherein the light sensitive component comprises at least one of a camera, an ambient light sensor, or a flicker sensor.

11. The electronic device of claim **8**, wherein the seal comprises a compressible foam.

12. The electronic device of claim **8**, wherein the seal extends from the top surface of the base support to the external surface of the housing.

13. The electronic device of claim **8**, further comprising a flexible printed circuit board disposed between the seal and the base support, the seal retaining the flexible printed circuit board adjacent the base support.

14. The electronic device of claim **8**, wherein at least one of the interior surface or the external surface comprises an air vent.

15. The electronic device of claim **8**, wherein the light emitting component is coupled to a top surface of the base support.

16. The electronic device of claim **15**, wherein:
- the base support comprises a ledge;
- the seal surrounds an upper portion of the base support; and
- the seal comprises a plastic shim contacting the ledge.

17. An electronic device, comprising:
- a base support;
- a dust seal coupled to the base support via an interference fit; and
- a flexible printed circuit board disposed between the dust seal and the base support, the dust seal retaining the flexible printed circuit board adjacent the base support.

18. The electronic device of claim **17**, wherein:
- the flexible printed circuit board is coupled to a light emitting device mounted to the base support;
- the dust seal encircles the light emitting device; and
- the dust seal resists light and dust.

19. The electronic device of claim **17**, wherein:
- the dust seal comprises a plastic shim and a foam portion, the plastic shim having a bottom surface coupled to the base support with a pressure sensitive adhesive and a top surface coupled to the foam portion with the pressure sensitive adhesive; and
- the foam portion comprises a compressible foam.

20. The electronic device of claim **19**, wherein:
- the base support comprises a ledge;
- the dust seal surrounds an upper portion of the base support; and
- the plastic shim contacts the ledge.

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